

Curriculum Vitæ

Antoine Kassis
aelkas.github.io

antoine.kassis@ens-paris-saclay.fr, +33 6 60 55 94 71
github.com/aelkas

Academic Background

- **Master 2, MVA (Mathématiques, Vision, Apprentissage)**, École Normale Supérieure Paris-Saclay 2026-2027
- **Masters 1 (M1) Mathematics Magistère** in fundamental mathematics, Université Paris-Saclay 2025-2026
- **Third-year (L3) Double Degree, Mathematics Magistère**, Université Paris-Saclay, graduated with highest honor ("mention très bien") 2024-2025
- **Second-year (L2) Double Degree in Mathematics and Computer Science**, Université Paris-Saclay 2023-2024
- **First-year (L1) Double Degree in Mathematics and Physics**, Université Paris-Saclay 2022-2023
- **Final Year of High School**, Antonine International School Ajaltoun, Lebanon
French Baccalaureate with specializations in Mathematics, Physics-Chemistry, and Expert Mathematics; graduated with Highest Honors 2021-2022

Research

- **Research internship (TER), Institut Mathématique d'Orsay / INRIA DataShape**
Supervised by Mr. Marc Glisse. Graded 19/20.
 - Generalized the curvature inequalities of D. Cohen-Steiner ("Inequalities for the curvature of curves and surfaces") from curves and surfaces to arbitrary dimension.
 - Main theorem bounds differences in Lipschitz-Killing curvatures by integrating the Euler characteristic over hyperplane sections.
 - Tools: persistent homology, integral geometry (Crofton, Steiner, Hadwiger's theorem), and Morse theory for manifolds with boundary.April – June 2026
- **Research Internship, Institut Mathématique d'Orsay**
Supervised by Mr. Yves de Laszlo.
 - Co-authored a comprehensive text on group representation theory, covering the progression from finite to solvable groups.
 - Proved foundational results, including the Lie-Kolchin theorem, and developed key illustrative examples for the text.March – May 2025

Professional Experience and Skills Acquired

- **Member of Automatants (CentraleSupélec)**
Active member of the student Artificial Intelligence association. Built guided deep learning projects: a GAN generating images of cats, a CNN for Alzheimer's detection, and image classifiers. Participated in team brainstorming for AI related hackathons and seminars, including discussions on geometric deep learning. Contributed to designing educational slides for introductory machine learning sessions. November 2025 – April 2026

- **Data Analysis Internship at TLD (ALVEST Group):**

Supervised by Mr. Frédéric Bouilhaget and Mr. Alexandre Dupaquis.

- Developed a data analysis methodology to diagnose failure causes from the error logs of electric aircraft tractors (TPX-200-MTX-E).
- Implemented temporal clustering algorithms (DBSCAN) to distinguish between root-cause errors and secondary effects within error bursts.
- Built two composite behavioural scores: a cause/effect axis from a Beta-law fit of normalized in-burst positions, and a recurrent/impulsive axis from the dispersion of appearance days. Introduced a rank-based position metric to handle timestamp collisions, validated against hierarchical clustering across seven time scales.
- Created a final interactive visualization tool (Python, Dash) to present a behavioral "Identity Card" for each error, facilitating intuitive diagnostics for engineers.

June – July 2025

- **UX Workshop at Johnson/Luke (for Dubai Municipality):**

Contributed to testing a revised information architecture for the Dubai Municipality website, with test results indicating a 25% increase in user clarity and efficiency. Acquired practical skills in user testing methodologies and UI/UX evaluation.

Aug 2024

- **Data Science Internship at Johnson/Luke:**

Applied NLP and ML techniques to analyze customer feedback on the mobile application of a Lebanese bank, in a context where the country's financial crisis had eroded consumer confidence. Identified key issues impacting user retention by structuring insights within a graph database.

July – Aug 2024

- **Tutor in Mathematics, Computer Science, and Physics with Eddmon:**

High school level.

2023-2024

Technical Skills

- **Programming Languages:** Python, C, C++, Java, OCaml
- **Databases:** PostgreSQL, Neo4j
- **Tools and Technologies:** L^AT_EX, Git, Hex
- **Languages:** French (fluent), English (C1/C2, TOEFL iBT: 108/120), Arabic (fluent)

Projects

- **Note:** A complete list of software and algorithmic projects is available on my GitHub: github.com/aelkas
- **Shaaer — Arabic Poetry Transformer (Python):**
A GPT language model trained from scratch on classical Arabic poetry, following the nanoGPT approach, with a custom grapheme-cluster-aware BPE tokenizer for Arabic. Reaches approximately 3.01 bits per character, below the bzip2 compression baseline.
- **Quad-trees (C):**
Binary images represented as quadtrees, with morphological operations and a compression pass that detects structurally identical subtrees anywhere in the image and shares them, turning the tree into a directed acyclic graph so that repeated patterns are stored once.

- **Graphs and Logic (Python):**

Object-oriented tools for directed open graphs, extended into Boolean circuits with adders and matrix operations. Gate transformations are delegated to specializations of an abstract node class, so circuit simplification and evaluation become graph rewriting.

- **Colt Express (Java):**

Full digital implementation of the board game Colt Express with a graphical interface: rules engine, game state, and the programmed-action turn system.

- **Maze Generation and Solving (OCaml):**

A maze model with several generation and solving algorithms and an animated HTML view, each module split into interface, implementation, and tests.

- **Computation and Precision (C):**

Exploration of programming limits by implementing sequences such as Ackermann's and testing boundary conditions related to Euler's number, plus linked-list routines enumerating all permutations of n with as few allocations as possible.

Seminars and Schools Attended

- **DataShape seminars**, INRIA Saclay — attended on the recommendation of Nina Otter 2026
- **Summer school on percolation**, Institut des Hautes Études Scientifiques
Under Thomas Hutchcroft 2025
- **Maths en herbe**, Institut des Hautes Études Scientifiques January 15, 2025
- **Mathematical Culture**, Institut Mathématique d'Orsay, Paris-Saclay
November 14-18 and December 9, 2024
- **Philosophy and Mathematics Seminars**, École Normale Supérieure PSL
Bi-monthly participation on Mondays in 2023-2024

Interests

- Computational topology and geometry, persistent homology, topological and homological algebra, representation theory, geometric deep learning
- Sports
- Violin
- Algorithms and puzzle solving